VULTR: AGENT RUSH HACKATHON · TRACK 01: BLAST RADIUS ZERO

BlastRadius Zero

Containment-First Multi-Agent Cloud Execution Gatekeeper & Ephemeral gVisor Sandbox Verifier on Vultr Cloud Compute and Vultr Serverless Inference.

Centralized Control Layer on Vultr

Intercepts autonomous agent remediation plans (Shell, SQL, Terraform, Kubernetes), enforces deterministic policy checks, and dispatches verified work into ephemeral gVisor (runsc) sandboxes on Vultr Cloud Compute.

Live Deployment & Repository

Public Demo: https://aariv.me/blast-radius-zero/
GitHub Repo: https://github.com/draarivpatel-ui/blast-radius-zero
Team: BlastRadiusZero (Aariv Patel)

THE PROBLEM

Why Autonomous Cloud Agents Cause Catastrophic Outages

When SRE and DevOps agents operate without deterministic containment, a single hallucinated command or prompt injection can wipe production state or trigger runaway GPU billing.

1. Runaway Recursive Subagent Loops

Cyclic tool calls spin up 48x GPU instances ($14,200/hr) and issue Redis FLUSHALL during peak traffic.

2. Destructive Schema & IAM Drift

Unbounded DROP COLUMN migrations paired with iam:* and AdministratorAccess privilege escalation.

3. Supply-Chain Prompt Injection

Untrusted PR descriptions trick triage agents into running curl $(env | base64) to exfiltrate credentials.

4. Thundering-Herd Cache Stampedes

Deleting HAProxy configs or flushing caches under BGP route leaks collapses downstream databases.

CONTAINMENT ARCHITECTURE

Vultr Cloud Compute gVisor Sandbox & Serverless Inference

Containment Pillar	Enforced Mechanism on Vultr	Verification Receipt
1. Process Isolation	gVisor (runsc) container runtime with --network=none & seccomp-bpf filter	**Host runtime untouchable**
2. Secret Hygiene	env_mounted=0; zero API keys or cloud credentials inside sandbox	**secrets_exposed=0**
3. Resource Limits	cgroups v2 hard caps: 1.0 vCPU, 512 MB RAM, pids.max=64, 15,000ms cap	**wall_time=42ms, rss=38.4MB**
4. Lifecycle Discipline	Ephemeral rootfs destroyed after every task with SHA-256 stdout receipt	**DESTROYED_CLEAN (18ms)**

```
[VULTR-CONTROL-PLANE] container_id=vultr-sbx-ewr-a4054829c7 runtime=runsc(gVisor) isolation="--network=none --read-only --pids-limit=64 --memory=512m" [CONTAINMENT-GATE] HTTP 403 BLOCKED raw agent proposal (3 BLOCK, 1 WARN, blast_score=100/100) -> synthesized Zero-Blast-Radius plan [VERIFIER-RECEIPT] exit_code=0 wall_time=42ms secrets_exposed=0 lifecycle=DESTROYED_CLEAN
```

DETERMINISTIC POLICY GATEKEEPER

Four Hard Policies + Zero-Blast-Radius Remediation Synthesis

POL-01 · No Unbounded Deletion

Blocks rm -rf, DROP TABLE/COLUMN, TRUNCATE, DELETE without WHERE, FLUSHALL, kubectl delete --all, terraform destroy.

POL-02 · Least Privilege IAM

Blocks iam:*, AdministratorAccess, Resource "*", and wildcard sts:AssumeRole escalations.

POL-03 · Cost Cap <= $50/hr

Blocks runaway GPU cluster scaling bursts, replica surges (>20 replicas), and hourly spend spikes above $50/hr.

POL-04 · Zero Secret Exfiltration

Blocks curl/wget with environment variables, /proc/self/environ reads, and base64 credential exfiltration.

VERIFICATION & AUDIT EXPORTS

Shadow Canary Simulation & Cryptographic Audit Bundles

Every intercepted incident produces verifiable before/after diffs, P95 latency & error recovery curves, and one-click compliance exports.

4 Built-In Attack Scenarios + Live Composer

- 4/4 destructive agent proposals contained (HTTP 403 BLOCKED)
- Synthesizes safe, bounded remediations with Blast Radius Score = 0/100
- Interactive cloud topology SVG highlights impacted services in real time

Production Audit Artifacts

- **Signed Audit JSON:** Full policy state, line-level findings, SHA-256 receipt
- **Markdown Report:** Incident post-mortem ready for PR review
- **SARIF 2.1.0 Bundle:** Direct import into GitHub Code Scanning